

ITAO5 - Activity.

1.) CCTV image contains Salt - and - pepper noise.

* Technique: median Filter.

* Reason: It effectively removes Salt - and - pepper noise while preserving image edges.

2.) ~~sat~~ Satellite image is corrupted by Gaussian noise.

* Technique: Gaussian Filter

* Reason: It reduces Gaussian noise by smoothing the image and improving overall quality.

3.) medical x-ray image has random noise, but fine edges must be preserved.

* Technique: Bilateral Filter.

* Reason: It removes random noise while preserving important edges and fine details in the x-ray image.

4) A Scanned document has black and white dots scattered randomly.

* Technique: median filter.

* Reason: It removes black and white (salt-and-pepper) noise while keeping text edges clear.

5) A low-light image shows heavy grain without clear boundaries.

* Techniques: Gaussian filter

* Reason: It reduces grain and produces a smoother image.

6) A foggy road image has very low contrast.

* Technique: Histogram Equalization

* Reason: It enhances image contrast, making objects and roads more visible.

7) An underwater image appears dull and washed out.

- * Technique: Contrast stretching

- * Reason: It improves brightness and color contrast, making the image clearer and more vibrant.

8) A chest x ray shows poor visibility of internal structures.

- * Technique: Histogram Equalization

- * Reason: It improves contrast and makes internal structures more visible.

9) A night-time traffic image lacks intensity variation.

- * Technique: Contrast stretching

- * Reason: It expands the intensity range, making dark and bright regions clearer.

10) A gray scale image uses only a narrow intensity range.

- * Technique: Contrast stretching

- * Reason: It increases the dynamic range and enhances the overall contrast.